

Supplementary Material for IAD-Risk

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1 Scope

This supplementary material documents the reproducibility boundary for the manuscript “IAD-Risk: Risk-Aware Identity-Agenda Disentanglement for Scholarly Work Deduplication.” It is intended to accompany the main manuscript, not to replace the project repository or the final artifact release. The repository contains executable code, fixture data, schema contracts, and build scripts; large third-party raw data, model checkpoints, and full experimental outputs are kept outside Git and must be distributed through an artifact release when numerical result reproduction is required.

2 Reproduction Levels

Table 1: Reproduction levels and expected evidence.

Level	Required input	Evidence produced
L0	Source tree and configured Python environment	CLI help, public-release scan, unit tests, and PDF build logs.
L1	Public miniature fixtures in tests/fixtures	Eval-source files and an IAD-Bench fixture package with the documented schema.
L2	Independently obtained public raw files	Rebuilt DeepMatcher, SciRepEval-style, OpenAlex-style, and OpenCitations-derived evaluation sources.
L3	Released artifact package	Fixed tables, predictions, logs, manifests, checksums, and commit identifiers for result audit.

3 IAD-Bench Schema Contracts

The main manuscript summarizes the schema in prose. The full tables below define the required document and pair fields used by fixture rebuilding, public-source rebuilding, and artifact validation.

Table 2: IAD-Bench document schema contract. Missing values are represented by empty strings, empty arrays, or null values rather than by dropping required fields.

Field group	Required fields	Interpretation role
Record identity	<code>document_id</code> , <code>source_dataset</code>	Makes each standardized record traceable to its originating public source.
Text and authorship	<code>title</code> , <code>abstract</code> , <code>authors</code>	Provides the textual and author evidence used by identity and agenda features.
Bibliographic metadata	<code>year</code> , <code>venue</code> , <code>doi</code> , <code>arxiv_id</code> , <code>openalex_work_id</code>	Supplies identifier and publication-context evidence for same-work decisions.
Agenda context	<code>topics</code> , <code>references</code>	Supplies topical and citation evidence used for same-agenda and hard-negative construction.

Table 3: IAD-Bench pair schema contract. The schema records identity labels, agenda labels, provenance, split assignment, and hard-negative status for each pair.

Field group	Required fields	Interpretation role
Pair identity	<code>pair_id</code> , <code>source_document_id</code> , <code>target_document_id</code>	Makes every row traceable to two standardized documents and prevents anonymous aggregate-only reporting.
Relation targets	<code>relation_label</code> , <code>expected_label</code> , <code>expected_agenda_label</code>	Separates same-work identity from agenda relatedness before metrics are computed.
Evidence source	<code>label_source</code> , <code>label_strength</code> , <code>label_provenance</code>	Records whether a label is gold, proxy, silver, or human-audit evidence and what signals support it.
Evaluation control	<code>split</code> , <code>hard_negative_level</code>	Controls train/dev/test/audit use and identifies agenda-level hard negatives for HNFMR.

4 Environment Setup

The repository is designed to run from a standard Python environment without committing raw third-party data. Reviewers can validate the code and fixture-level reproduction path from the project root with the following commands:

```
conda create -n iad-sieve python=3.11 -y
conda activate iad-sieve
python -m pip install -e .
python -m iad_sieve.cli --help
python scripts/check_public_release.py
python manuscript/scripts/verify_fixture_rebuild.py
```

The environment check does not download full raw datasets. It verifies package installation, CLI availability, public-release boundaries, and no-network fixture rebuilding. Full numerical result reproduction still requires the L2/L3 data and artifact requirements described below.

5 No-Network Fixture Rebuild

The fixture rebuild validates the data adapters and IAD-Bench schema without depending on external network access. It is not a substitute for the full paper experiment.

```
python -m iad_sieve.cli prepare-deepmatcher \  
  --table-a tests/fixtures/deepmatcher/tableA.csv \  
  --table-b tests/fixtures/deepmatcher/tableB.csv \  
  --pairs tests/fixtures/deepmatcher/test.csv \  
  --dataset-name deepmatcher_fixture \  
  --output-dir outputs/repro_fixture/deepmatcher  
  
python -m iad_sieve.cli prepare-scirepeval-proximity \  
  --metadata tests/fixtures/scirepeval/metadata.jsonl \  
  --pairs tests/fixtures/scirepeval/scidocs_cite_pairs.csv \  
  --dataset-name scirepeval_fixture \  
  --output-dir outputs/repro_fixture/scirepeval  
  
python -m iad_sieve.cli prepare-openalex-weak-labels \  
  --works tests/fixtures/openalex/works.jsonl \  
  --citations tests/fixtures/openalex/coci.csv \  
  --dataset-name openalex_fixture \  
  --output-dir outputs/repro_fixture/openalex \  
  --min-shared-references 1 \  
  --max-pairs 20  
  
python -m iad_sieve.cli build-iad-bench \  
  --source-dirs \  
    outputs/repro_fixture/deepmatcher \  
    outputs/repro_fixture/scirepeval \  
    outputs/repro_fixture/openalex \  
  --output-dir outputs/repro_fixture/iad_bench \  
  --seed 42
```

The manuscript package includes a verification wrapper for the same no-network path:

```
python manuscript/scripts/verify_fixture_rebuild.py
```

The wrapper runs the four fixture commands in a temporary directory and checks that each source adapter writes `eval_documents.jsonl`, `eval_pairs.jsonl`, and `dataset_summary.jsonl`, and that the assembled IAD-Bench package writes pair, document, split, summary, provenance, and dataset-card files.

6 Public-Source Rebuild

For full data reconstruction, raw public files should be obtained from their original sources and placed under local `data/raw/` directories. These directories are intentionally excluded from Git. The project keeps conversion commands and schema contracts under version control, while source licenses and access conditions remain governed by the original providers.

```
python -m iad_sieve.cli prepare-deepmatcher \  
  --table-a data/raw/deepmatcher/DBLP-ACM/tableA.csv \  
  --table-b data/raw/deepmatcher/DBLP-ACM/tableB.csv \  
  --pairs data/raw/deepmatcher/DBLP-ACM/test.csv \  
  --dataset-name deepmatcher_fixture \  
  --output-dir outputs/repro_fixture/deepmatcher
```

```

--table-b data/raw/deepmatcher/DBLP-ACM/tableB.csv \
--pairs data/raw/deepmatcher/DBLP-ACM/test.csv \
--dataset-name DBLP-ACM \
--output-dir data/processed/eval_sources/deepmatcher_dblp_acm

python -m iad_sieve.cli prepare-scirepeval-proximity \
--metadata data/raw/scirepeval/metadata.jsonl \
--pairs data/raw/scirepeval/scidocs_cite_pairs.csv \
--dataset-name scidocs_cite \
--output-dir data/processed/eval_sources/scirepeval_scidocs_cite

python -m iad_sieve.cli prepare-openalex-weak-labels \
--works data/raw/openalex/works_cs_sample.jsonl \
--citations data/raw/opencitations/coci.csv \
--dataset-name openalex_cs_sample \
--output-dir data/processed/eval_sources/openalex_cs_sample \
--min-shared-references 1 \
--max-pairs 5000

python -m iad_sieve.cli build-iad-bench \
--source-dirs \
    data/processed/eval_sources/deepmatcher_dblp_acm \
    data/processed/eval_sources/scirepeval_scidocs_cite \
    data/processed/eval_sources/openalex_cs_sample \
--output-dir data/processed/iad_bench/open_v3 \
--train-ratio 0.8 \
--dev-ratio 0.1 \
--seed 42

```

7 Public-Source Rebuild Audit Boundary

The L2 public-source rebuild should be treated as auditable only when the raw-input boundary, processing run, and derived-output boundary are recorded together. A reviewer does not need the raw third-party files in Git, but should be able to inspect a release-level `source_input_manifest`, a `processing_run_log`, and output summaries that identify which public files, commands, seeds, software versions, and checksums produced each derived evaluation source.

Table 4: Audit boundary for L2 public-source rebuilds. The table states the files that should accompany a rebuild when raw third-party data are not committed to Git.

Audit item	Required content	Audit use
<code>source_input_manifest</code>	Source name, acquisition date or version, original provider, local file name, record count when known, license boundary, and SHA256 checksum for each input file.	Shows that the rebuild used identifiable public inputs without redistributing those inputs in Git.
<code>processing_run_log</code>	Command line, code commit, environment summary, random seed, start and finish timestamps, input manifest reference, output path, and exit status for each processing stage.	Shows that DeepMatcher, SciRepEval-style, OpenAlex/OpenCitations, and IAD-Bench assembly commands were run under a fixed source revision.
Output summaries	Dataset summaries, split summaries, provenance summaries, row counts, label-strength counts, and checksum entries for derived files.	Lets reviewers compare derived evaluation artifacts against the manuscript scope before using them as numerical evidence.
Exclusion record	Explicit statement that raw third-party files, credentials, caches, model checkpoints, and local experiment outputs are excluded from the public Git package.	Prevents a clean repository from being mistaken for an incomplete or hidden data release.

The artifact release template therefore requires `source_input_manifest` and `processing_run_log` in addition to result tables, predictions, threshold logs, and split summaries. These files do not upgrade fixture-level reproduction into a full numerical audit by themselves. They provide the chain of custody needed to decide whether an L2 rebuild is consistent with the L3 result artifacts used to support manuscript numbers.

8 Data Processing Traceability

Table 5 gives the audit trace from public source files to derived evaluation artifacts. The repository commits the processing commands and schema checks, but keeps the raw source files and large derived outputs outside Git. This separation allows an independent reader to rebuild the same artifact classes from obtained public sources while respecting source licenses and keeping the public repository small.

Table 5: Traceability from public data sources to derived IAD-Bench artifacts. Git contains commands, schemas, tests, and manifests; full-size inputs and outputs are verified through external artifact checksums.

Stage	Local input boundary	Processing entry point	Expected output	Git boundary
DeepMatcher gold	Raw benchmark tables and pair labels.	prepare-deepmatcher.	Eval documents, eval pairs, and dataset summary.	Raw tables excluded; fixtures tested.
SciRepEval proxy	Public metadata and citation-pair files.	prepare-scirepeval-proximity.	Proxy agenda pairs and source summaries.	Full metadata excluded; schema retained.
OpenAlex weak labels	OpenAlex works and OpenCitations COCI files.	prepare-openalex-weak-labels.	Silver hard negatives with provenance.	Large sources excluded; rules retained.
IAD-Bench assembly	Processed source directories from previous stages.	build-iad-bench.	Unified documents, pairs, splits, provenance, and dataset card.	Full package released as L2/L3 artifact.
Result audit	Prediction, score, threshold, log, and metric files.	Artifact finalization and validation scripts.	Manifest, checksums, tables, predictions, and command logs.	External artifact required.

9 Artifact Package Requirements

Full numerical result reproduction should use a released artifact package rather than raw Git-tracked files. A complete artifact package should contain a manifest, checksums, tables, predictions, configuration files, logs, and the commit identifier used to generate the results.

```
paper-artifacts/
  README.md
  manifest.json
  checksums.sha256
  configs/
  tables/
  predictions/
  reports/
  logs/
```

The manifest should record the source data version, processing commands, dependency versions, random seeds, thresholds, split configuration, hardware summary, and SHA256 checksum for every result file. The checksums should be validated before using artifact files to support claims in the manuscript.

```
python manuscript/scripts/build_artifact_release_skeleton.py \
  --output-dir /path/to/release \
  --repository-commit <commit>
python manuscript/scripts/populate_artifact_release.py \
  --artifact-dir /path/to/release \
```

```

--source-dir /path/to/source-artifacts \
--preflight-only
python manuscript/scripts/populate_artifact_release.py \
--artifact-dir /path/to/release \
--source-dir /path/to/source-artifacts
python manuscript/scripts/finalize_artifact_release.py \
--artifact-dir /path/to/release
python manuscript/scripts/validate_artifact_release.py \
--artifact-dir /path/to/release

```

The release preparation commands create a scaffold, check source artifact completeness before copying, copy generated result files from a source artifact directory, refresh manifest and checksum metadata, and then validate the final package. The `--preflight-only` command reuses the release manifest copy plan to check required source artifact paths and optional mapping paths without copying files, writing `artifact_population_log.jsonl`, or running finalization. A passing source preflight shows that the source artifact directory is structurally ready for population; it does not prove row-level schemas, checksums, public accessibility, source-control commit binding, or numerical correctness. The validation command checks the manifest schema, required artifact files, required result identifiers, checksum coverage, repository commit metadata, claim-boundary flags, conditional claim artifacts, and exclusion of raw third-party data, caches, credentials, local outputs, and model checkpoints.

The main Open-v2 result artifact is `open_v2_main_results`. Its CSV file must include per-row denominator counts, the per-row threshold source, the scope label used in the main table, automatic merge count, block count, defer count, automatic merge coverage, defer rate, and capacity-normalized review load. This row-level schema lets reviewers trace each reported same-work F1, FMR, and HNFMR value to the evaluated pair scope, denominator, operating-point source, selective-decision coverage, workload boundary, and fixed artifact version.

The prediction and threshold artifacts must be row-auditable rather than opaque model dumps. The `iad_risk_predictions` JSONL file must include system, `pair_id`, `source_document_id`, `target_document_id`, expected labels, label strength, hard-negative level, split identifiers, relation-head scores, `work_threshold`, `agenda_block_threshold`, `risk_threshold`, threshold source, and `merge_prediction`. The `representation_baseline_scores` JSONL file must include the same pair and label fields plus normalized score, `score_field`, `threshold_value`, threshold source, and `merge_prediction`. The `supervised_baseline_predictions` JSONL file must include the same pair and label fields plus `match_probability`, `threshold_value`, threshold source, and `merge_prediction`. The `threshold_selection_logs` JSONL file must include system, `threshold_name`, `threshold_value`, `selection_split`, `selection_metric`, `selection_rule`, `applied_scope`, and `score_field` so that fixed operating points cannot be selected from the test rows after evaluation.

The artifact manifest uses conditional claim flags so that optional stronger claims cannot be enabled without their evidence files. Confidence-interval claims require `bootstrap_intervals`; component-causality claims require `ablation_suite`; human-validation claims require `manual_validation_slice`; threshold-stability claims require `threshold_sensitivity_grid`; cluster-level quality claims require `cluster_metric_summary` and `cannot_link_audit`; and broad method-ranking claims require the interval, validation, and threshold-grid artifacts together. If a conditional flag is set to true, the corresponding artifact row must be marked required and must include a file path and SHA256 checksum.

Cluster-level quality claims require evidence beyond pair-level prediction files. The `cluster_metric_summary` artifact should include cluster assignments, cluster contamination rate, singleton rate, merge coverage, conflict counts, and a pair-to-cluster trace from accepted links to connected

components. The `cannot_link_audit` artifact should report cannot-link coverage, violations, unresolved conflicts, and identifier-conflict rules under the same threshold and merge policy used for the reported results. Without both files and their checksums, the manuscript should discuss pair-level false-merge control only and should not claim cluster-level contamination reduction.

10 Baseline Audit Boundary

The main manuscript reports the baseline decision rule in prose to keep the result narrative readable. The full audit matrices below define which baseline families enter the primary result table and which comparison controls must hold before a broad ranking claim is made.

Table 6: Baseline inclusion rationale. A baseline enters the primary result table only when it has auditable outputs under the reported pair scope.

Baseline family	Current role	Inclusion or exclusion rule
Exact identifier matching	Important sanity-check family for DOI, arXiv, and publication-lineage evidence.	Excluded from primary result table when only utility code or fixture-level checks are available; full metrics require prediction files and checksums.
Title-normalization rules	Useful lower-bound family for metadata-only deduplication.	Excluded until the same pair scope, threshold records, metric summaries, and error cases are released.
Traditional entity-resolution systems	Relevant workflow baselines for blocking, matching, and clustering comparisons.	Included as primary evidence only when metric summaries, prediction files, threshold records, and checksums are available.
Scientific representation baselines	Primary stress-test evidence for single-space scientific relatedness.	Included because SciNCL cosine and SPECTER2 adapter cosine have reported score summaries under Open-v2.
RoBERTa pair classification	Strong supervised comparator for pair-level matching.	Included as a primary supervised comparator while keeping the result below broad ranking claims.

Table 7: Baseline fairness controls for the Open-v2 evidence snapshot. The controls make the reported comparison auditable while preserving the scope boundary of each row family.

Control	Application in the reported table	Interpretation boundary
Input rows	Baselines use the same IAD-Bench pair records within each reported row scope.	A stricter ranking requires same-scope released prediction files for all rows.
Split use	Trainable rows use the same train/dev/test split field; non-trainable representation rows are reported as fixed score baselines.	Different available scopes should not be read as a single comparative ranking.
Threshold selection	Threshold-sensitive rows use validation-selected operating points or recorded fixed thresholds.	Post-hoc test-threshold selection is not used for the reported claims.
Feature boundary	Label source, provenance, and split identifiers are audit fields, not predictive features.	Provenance can explain evidence strength but cannot be used as a shortcut feature.
Metric implementation	All systems report same-work F1, FMR, and HNFMR from the same metric implementation.	Metric comparability does not remove the need for released predictions and checksums.

11 Claim-Evidence Matrix

Table 8 maps the major manuscript claims to the evidence required for responsible interpretation. The matrix is part of the submission boundary: a claim should be weakened or removed if the corresponding evidence file is absent from the released artifact package.

Table 8: Claim-evidence matrix for the main manuscript.

Claim	Required evidence	Boundary
Identity-agenda confusion is a practical false-merge risk.	Open-v2 hard-negative evaluation with FMR and HNFMR fields.	Supports a targeted scholarly deduplication failure mode, not a prevalence estimate for all domains.
IAD-Risk suppresses hard-negative false merges under the reported setting.	IAD-Risk prediction files, metric summaries, split identifiers, thresholds, and checksums.	Supports targeted false-merge suppression; does not establish broad method superiority.
IAD-Bench is provenance-aware.	Pair schema, label strength, label source, provenance fields, and fixture rebuild commands.	Supports a benchmark contract; does not turn silver or proxy labels into human gold.
Repository-level reproduction is possible without committing raw data.	L0/L1 fixture rebuild, public-release scan, unit tests, and documented public-source commands.	Verifies code and schema behavior; full numerical reproduction requires L2/L3 source files or artifacts.
Cluster-level quality is claimed only after pair-to-cluster audit.	<code>cluster_metric_summary</code> , <code>cannot_link_audit</code> , cluster assignments, cannot-link coverage, and cluster contamination rate.	Supports cluster-level quality claims only when the released artifact package includes these files with checksums; otherwise the claim remains pair-level.

12 Uncertainty and Ablation Requirements

Result packages that report confidence intervals should include bootstrap outputs generated from the exact prediction files used in the manuscript tables. The repository provides *run-iad-evidence-bootstrap* for stratified pair-level intervals and *run-bootstrap* for general metric intervals. A numerical interval should be cited only when the corresponding CSV file, command log, random seed, and checksum are present in the artifact package.

Threshold-stability claims should be tied to a released threshold-sensitivity grid generated from the same prediction files as the main tables. The grid should report threshold ranges, selected operating points, pair counts, same-work F1, FMR, HNFMR, random seeds, command logs, and checksums. Without this file, the manuscript should discuss only the fixed operating points reported in the main table.

Mechanism claims should be tied to explicit ablation outputs. The repository provides *run-iad-ablation-suite* for comparing full IAD-Risk decisions with variants that remove risk gating, agenda-non-identity blocking, or single-space separation. Without those outputs, the main manuscript should keep the mechanism claim at the level of stratified HNFMR evidence rather than claiming a complete ablation study.

Cluster-level claims should be tied to artifacts generated after the pair decisions are converted into connected components. A defensible package should include cluster assignments, a cluster metric summary, a cannot-link audit, and command logs for the merge policy. Without those outputs,

pair-level FMR and HNFMR should not be interpreted as proof that cluster-level contamination has been reduced or eliminated.

13 Reviewer Evidence Gate

This section states the evidence gate that should be applied before a reviewer reads the present results as stronger than the manuscript claims. The current package supports a bounded mechanism claim: identity-agenda separation and false-merge risk gating can suppress hard-negative automatic merges in the reported Open-v2 evidence snapshot. Stronger claims require released artifacts that bind the same rows, thresholds, logs, and checksums to the submitted text.

Table 9: Reviewer evidence gate for upgrading manuscript claims. Each stronger interpretation requires an explicit artifact source before it should appear in the abstract, conclusion, cover letter, or response to reviewers.

Review dimension	Current supported reading	Evidence required before stronger wording
Contribution clarity	Identity-agenda confusion is a targeted scholarly false-merge failure mode.	Broader task-replacement claims require source diversity, same-scope prediction files, and domain-facing error analysis.
Experimental strength	Open-v2 supports fixed-scope false-merge control under the reported operating points.	Model-ranking claims require same-scope prediction files, threshold logs, bootstrap intervals, and released metric tables.
Evaluation completeness	The manuscript separates gold, proxy, silver, and future manual evidence layers.	Component-causality or label-precision claims require an ablation suite, threshold grid, and manual-validation slice.
Method design soundness	Risk gating exposes identity, agenda, ANI, threshold, and cannot-link behavior.	Deployment claims require source-heldout validation, topic-heldout checks, review-load analysis, and cluster-level artifact audits.
Reproducibility	Git supports code-level checks, fixture rebuilds, schema validation, and artifact release preparation.	Full numerical audit requires an external release with manifests, checksums, command logs, and fixed commit identifiers.

The practical rule is a do not upgrade claims rule: do not strengthen the title, abstract, conclusion, cover letter, or reviewer response beyond the bounded Open-v2 mechanism claim unless the corresponding evidence row in Table 9 is satisfied by released artifacts.

14 Manual Validation Protocol

Manual validation is a future evidence layer for assessing label precision and difficult boundary cases; it is not completed in the current manuscript package. A defensible validation slice should sample 500–1,000 pairs across the strata most likely to affect claim strength: OpenAlex/OpenCitations silver hard negatives, high-score representation false-merge candidates, IAD-Risk blocked or deferred pairs, version-boundary cases, identifier conflicts, and sparse-metadata cases.

Table 10: Manual validation protocol required before stronger label-precision claims.

Protocol item	Requirement	Claim boundary
Sampling	Stratified 500–1,000 pair slice across silver hard negatives, model-disagreement pairs, version-boundary pairs, and sparse-metadata pairs.	The slice estimates difficult-case precision; it is not a prevalence estimate for all scholarly records.
Review process	two independent reviewers inspect titles, authors, venues, identifiers, abstracts, and citation context while blind to model scores and merge decisions.	Model outputs cannot determine the human label.
Label schema	Reviewers choose same work, agenda non-identity, unrelated, version boundary, or uncertain, and record a short evidence note.	Uncertain cases remain separate and are not forced into gold labels.
Adjudication	Disagreements are resolved by an adjudication log that records the final relation, rationale, and reviewer IDs or anonymous reviewer codes.	Adjudicated labels become auditable only when the log is released.
Reporting	The artifact package reports inter-annotator agreement, per-stratum counts, conflict rates, and checksums for the reviewed slice.	The manuscript must not claim human gold or stronger label precision until these files exist.

15 Claim Boundary

The main manuscript treats DeepMatcher-style labels as gold identity evidence. It treats SciRepEval and SciDocs-style proximity as same-agenda proxy evidence. It treats OpenAlex and OpenCitations-derived pairs as silver hard-negative evidence. OpenAlex and OpenCitations signals are not human gold labels. Cluster-level quality is not claimed unless the released artifact package includes cluster assignments, `cluster_metric_summary`, `cannot_link_audit`, `cannot_link` coverage, and cluster contamination rate. Broad method-ranking claims require stronger source-heldout evidence and manual validation beyond the current manuscript package.